

Ruko (रुको)

Stop before you pay.

Share a chat investment tip, hear the red flags in Hindi, pause, then verify on SEBI Check or follow the right recovery path.

SANGYAN Investor Resilience Hackathon · independent prototype

The problem

Ramesh, 46, runs a hardware shop in a tier-3 town in UP. Hindi first, low-end Android, patchy data.

Relatives added him to "stock tips" groups on WhatsApp and Telegram. A message promises **"guaranteed 20% a month"**, a **₹4999 VIP fee to a personal UPI ID**, and a **SEBI registration number, today only**.

He has never used SEBI Check or SCORES and does not know about 1930. He will not read a long screen, but he will listen to a short answer.

What Ruko does

1. **Share** the message from WhatsApp or Telegram to Ruko. Personal numbers are masked on the phone.
2. **Hear** the result in Hindi: band, class (promotion or education), the exact words that raised each flag, one analogy.
3. **Verify**: registration numbers become "couldn't verify", one tap to SEBI Check.
4. **Pause**: a three-line journal and "wait 24 hours". It stays on the phone.
5. **Get help**: already paid means 1930, then the bank, then cybercrime.gov.in, plus a complaint draft. For a broker problem, the SCORES steps.

It works offline too: the same rules run on the phone.

How Ruko decides

- **Rules decide the band**, from 19 signals in one shared file (`signals.v0.json`) read by the server and the phone.
- An optional LLM may only add tags and card text, behind a 3-second timeout and a circuit breaker.
- **Guardrail linter**, 8 checks on every response: no buy, sell, or hold; no tickers; no "safe" or "scam" verdicts; no predictions; only allowlisted links. It fails closed.
- **Never says "safe"**. Every result ends with "No flags does not mean it is safe to send money."

Architecture

```
WhatsApp / Telegram --share--> Ruko PWA ----masked text, HTTPS----> Spring Boot (stateless)
      | on-device rules + templates      api -> pipeline -> rules/band
      | journal, OCR, masking             -> explain (templates + optional LLM)
      |                                  -> guardrail linter
      | '<----- same JSON shape ----->' voice (Bhashini, optional)
      |                                   content (recovery, how Ruko decides)
```

- One container on JRE 21 serves both the PWA and the API, in 512 MB.
- Every external dependency has a fallback: LLM down means templates, Bhashini down means the phone's own voice, API down means on-device rules.

Results (80 labelled messages)

	Result	Target
Scams at some concern or higher	40 / 40	$\geq 90\%$
Genuine education flagged	0 / 25	$\leq 10\%$
Server and on-device agreement	80 / 80	100%
Linter: template outputs clean	320 / 320	100%
Server p95, LLM off	14 ms	≤ 1.5 s
Largest result, gzipped	0.8 KB	≤ 6 KB
Message text found in logs	0 of 1407 lines	0

The fixtures are also the development set, so these numbers are optimistic. 9 known gaps are listed in `eval-report.md`.

Privacy and guardrails

- **Message text lives for one request.** It is never written to disk, logs, or the voice cache. A canary log audit runs in CI.
- **The journal never leaves the phone.** There is no journal API route, and a browser test enforces it.
- **Permissions:** share target; clipboard and photo only on tap; notifications optional. No camera, location, or contacts.
- **Links:** only `siportal.sebi.gov.in` , `scores.sebi.gov.in` , `cybercrime.gov.in` , and `tel:1930` .
- **SEBI list:** there is no public registration-number list, so Ruko deep-links to SEBI Check.

DPDP principles applied: notice, consent, purpose limitation, minimisation. This is not legal advice.

Behaviour study

Protocol ready: 8–10 consenting adults, 10 fixed messages, with and without Ruko. It measures correct identification (target +30 points) and restating the reason (target $\geq 80\%$).

Sessions run so far: n = 0. No behaviour numbers are claimed. Results come only from real sessions (`docs/study/results.csv` , scored by `scripts/score-study.mjs`).

Limits and next

- Hindi and English. **Next language: Marathi** (named, not built).
- Spoken input (ASR) is built behind a flag and off.
- Domain-age lookup (S10) is off.
- Known rule gaps: 9 of 80 fixtures, each pinned by a test.
- Text-path speed was measured on an emulated slow 3G phone; a real low-end Android is still needed.

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Live: ruko-187c.onrender.com · **API:** ruko-187c.onrender.com/swagger-ui.html · **Code:** github.com/SunilMaurya-18/Ruko

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